

REVISED ARCHITECTURE · FOUR ROOMMATES ONLY

Shared Flat OS

A simple mobile-first web / PWA for one college flat. Tap who you are. Log milk, vegetables, groceries and bills. See who is home via flat Wi-Fi. Rahul must approve when someone says they paid him.

Ankit Joshi

Jayash Ghelot

Rahul Verma

Lakshit Pareek

Users

Exactly these four. No public signup.

Login

Tap your name. Phone remembers you.

Presence

Home Wi-Fi beacon — not GPS

Money

Integer paise. Settlement needs receiver approval.

V3

Removed. No AI / prediction / voice.

Monthly report

Now in V1

Date

10 August 2026

Version

1.1 — simplified after review

This revision replaces v1.0. No V3. Monthly report, vegetables, master list, and approve-my-payment are V1. Login is only “Who are you?”. Presence uses home Wi-Fi, not background GPS. This is not a bank — do not over-secure a four-person flat.

What changed in this revision

Topic	Old plan	Now
V3	AI, prediction, voice later	Deleted. Not in the product.
Monthly report	V2	V1
Vegetables	Generic groceries	Own add flow + type segregation
Login	Google / Apple / magic link	Tap Ankit / Jayash / Rahul / Lakshit
Master list	Activity log (vague)	Date, time, who, type, what, price — filterable diary
Settlement	One person marks paid	Payer claims → receiver Approves or Rejects
Presence	Foreground GPS / later native app	Home Wi-Fi: who is connected to the flat router
Security	Enterprise-style RLS / OAuth	Four trusted people. Correct money math. No login maze.

Contents

1. A. Product architecture
2. B. Feature map (V1 + tiny later list)
3. C. User flows
4. D. Presence — home Wi-Fi
5. E. Database
6. F. API
7. G. Who you are
8. H. UI / UX
9. I. Simple rules (not a bank)
10. J. Stack
11. K. Phases

A. Product architecture

One flat. Four people who already trust each other. One website they can install as a PWA. It should answer everyday questions without WhatsApp maths:

- Who is at the flat?
- Who paid for milk / sabzi / bill?
- What was added, when, by whom, at what price?
- Who owes whom — and has the other person confirmed the cash/UPI?
- What did we spend this month?
- Whose chore / who has the key / what is broken?

System shape

Four phones (browser or installed PWA)

| tap your name once



Shared Flat OS website



Simple backend + database

├ members (the four names)

├ expenses + master list

├ settlements (claim → approve)

├ monthly totals (computed)

└ presence (updated by home Wi-Fi beacon)

Home Wi-Fi

└ tiny always-on beacon (old phone / laptop / Pi / router)

sees which roommate phones are connected

→ Ankit At Flat, Rahul Away, ...

Rules that stay (these are correctness, not “security theater”)

Rule	Why
Money in integer paise	₹640 / 3 must not drift
Creating an expense ≠ paying a debt	Bill and cash are different moments
Settlement needs receiver approval	Stops “maine de diya tha” fights
Master list never lies	Every add has date, time, who, type, what, price
Types to segregate	Milk / Vegetables / Groceries / Food / Household / Bills / Other
No V3, no chat, no UPI auto-pay, no GPS tracking	Stay a flat OS, not a startup

B. Feature map

V3 is removed

No consumption prediction, no AI receipt split, no voice commands, no anomaly engine, no “smart insights.” If it is not useful this month for four roommates, it is not in the product.

V1 — this is the product

Area	What you get
Who are you	Four name buttons. This phone remembers you. Switch name from a small link.
Home	Who is home, you owe / you receive, veg + milk peek, due alerts, today's chore
Fast add	₹ amount + Type chip + title + paid by + split everyone / selected / custom → Save
Vegetables	Dedicated “+ Veg” : sabzi today, optional names (tomato, bhindi), price, who bought, who shares
Milk	Packets + price + who bought + who drinks milk (someone can opt out)
Segregation	Every add is typed: Milk, Vegetables, Groceries, Food, Household, Bills (rent/electricity/Wi-Fi), Other. Filter everywhere.
Master list	Diary of everything: date, time, who added, type, what, price. Filter by person / type / month.
Balances	Who owes whom after expenses minus <i>approved</i> settlements
Settle	“I paid Rahul ₹400” → Waiting for Rahul → Approve (clears) or Reject (still owed). Partial OK.
Monthly report	August total, spend by type, each person paid vs share, outstanding, approved settlements
Shopping + inventory	Need list vs what is in the flat (milk, veg, detergent...)
Recurring / rent / electricity	Wi-Fi & rent templates; electricity amount typed each month
Keys, chores, issues, flat info	Simple ops — same as before, no extra chrome
Presence	Phones on home Wi-Fi = At Flat. Planned away. UNKNOWN if no signal.

Later — tiny V2 only (optional, not promised)

Maybe later	Still never (old V3)
Guests, auto-skip milk when someone is away, receipt photo	AI, voice, prediction, chat, UPI rails, cameras, App Store GPS app unless Wi-Fi beacon fails badly
Push reminders if iPhone Home Screen PWA allows it	

C. User flows

Open the app (login)

```
Open site / PWA
→ already chose a name on this phone? → Home
→ else:
```

```
WHO ARE YOU?
```

```
[ Ankit Joshi ]
[ Jayash Ghelot ]
[ Rahul Verma ]
[ Lakshit Pareek ]
```

```
Tap once. Done. Nothing else.
```

No Google. No Apple. No email. No OTP. Optional later: one shared 4-digit flat code *only if* the link leaks outside. Default is off.

Add vegetables

```
+ Veg
  ₹180
  Tomato, bhindi (optional extra lines)
  Paid by: Ankit (default: me)
  Split: Everyone
  [Save]

→ Master list row:
  10 Aug 2026, 6:05 PM | Ankit | Vegetables | Sabzi | ₹180
```

Add milk / grocery / bill

Same sheet. Only the **Type** chip changes (Milk / Groceries / Food / Household / Bills / Other). Common case stays a few seconds.

Master list

MASTER LIST			filter: [All] [Veg] [Milk] [Ankit] [Aug]	
10 Aug 11:20	Rahul	Milk	2 packets	₹120
10 Aug 18:05	Ankit	Vegetables	Tomato, bhindi	₹180
10 Aug 20:12	Jayash	Bills	Electricity	₹4,380
11 Aug 09:40	Lakshit	Groceries	Oil + detergent	₹490

This is the screen people open during “bhai kisne pay kiya?” arguments.

Monthly report (V1)

AUGUST 2026

Total flat spending ₹42,840

Vegetables	₹4,120
Milk	₹2,100
Groceries	₹5,840
Electricity	₹3,240
Rent	₹24,000
Wi-Fi	₹799
Other	₹2,741

	Paid	Share	Net	
Ankit	₹12,100	₹10,710	+₹1,390	(should receive)
Jayash	₹8,200	₹10,710	-₹2,510	
Rahul	₹14,400	₹10,710	+₹3,690	
Lakshit	₹8,140	₹10,710	-₹2,570	

Approved settlements this month: ...

Still waiting for approval: Ankit → Rahul ₹400

Pay a debt (claim + approve)

Ankit: I paid Rahul ₹400
note: UPI (optional)
→ status: WAITING FOR RAHUL

Rahul sees on Home / Money:
Ankit says he paid you ₹400
[Approve] [Reject]

Approve → debt cleared, master list: "Settlement approved"

Reject → still owed, Ankit can try again

Partial: Ankit paid ₹200 of ₹400 → after approve, ₹200 still owed.

Only Rahul can approve a payment *to Rahul*. Ankit cannot mark it done alone. That matches real roommate fights.

D. Presence — can home Wi-Fi mean someone is home?

Yes — that is the right idea for this flat. GPS is the wrong idea.

If a roommate's phone is connected to the flat Wi-Fi, they are almost certainly at home. We should use that. We should not build background GPS tracking.

Honest limit (do not skip this)

The **website running on the phone cannot read the Wi-Fi name** ("JioFiber-5G" vs college Wi-Fi). iOS Safari, iOS PWA, Android Chrome, Android PWA — all block SSID / BSSID for privacy. So this will *not* work:

```
// this is impossible in a browser
if (wifiName === "Flat-WiFi") markAtHome();
```

Detection has to happen **from inside the home network**: "which known phones are currently connected to our router?"

How we will do it (simple, for four people)

Method	How	Automatic?
Home beacon (best)	A tiny always-on thing on flat Wi-Fi — old phone left charging, laptop, Raspberry Pi, or a script on the router if it allows. Every few minutes it looks at connected devices and tells the app: Ankit's phone is here, Rahul's is not.	Yes — nobody opens the app
Open-app ping	When someone opens the PWA, the site tries a local URL that only exists on home Wi-Fi (e.g. <code>http://192.168.0.50:8787/here</code>). If it succeeds, that person is At Flat.	Only when the app is opened
Planned away	"Rahul home 20 Aug–10 Sep" still shows away even if the phone is off.	Manual calendar
Fallback tap	Small "I'm home / I'm out" only if beacon is down. Not the daily ritual.	Manual

One-time phone setup (needed so Ankit ≠ "some iPhone")

Modern phones hide their real identity on Wi-Fi (Private Wi-Fi Address / randomized MAC). For *this home network only*, each roommate should:

- **iPhone:** Settings → Wi-Fi → (i) next to flat Wi-Fi → **Private Wi-Fi Address** → **Off** (or Fixed, then we save that MAC once).
- **Android:** Wi-Fi → flat network → Privacy → **Use device MAC**.
- Then in People → My devices → "This is Ankit's phone" (beacon records the MAC / name once).

After that the beacon can tell Ankit vs Rahul. Turning this off only for home Wi-Fi is normal and safe inside your own flat.

What can go wrong (we show UNKNOWN, we do not guess)

Situation	App shows
Phone on flat Wi-Fi	At Flat
Phone on mobile data / college Wi-Fi	Away (or Unknown if we never saw them leave)

Situation	App shows
Wi-Fi off / battery saver killed Wi-Fi / phone dead	Unknown — “signal unavailable”, not fake Away
Phone left at flat, person went to college	At Flat (same lie GPS would tell). Accept it, or they tap I’m out.
Guest uses Wi-Fi	Not a roommate unless we map that device
Wi-Fi reaches the building gate	May show At Flat while just outside — good enough for a flat
Beacon unplugged	Unknown for everyone until it is back; open-app ping still works

We will not do

Background GPS, geofences in the PWA, storing location trails, or claiming the browser can see your SSID. Wi-Fi-from-the-router is the honest automatic path for four people in one home.

E. Database (simple tables, correct money)

Table	What it stores
members	The four people: id, name. This is login.
expenses	type, title, amount_paise, paid_by, created_at, notes, split method
expense_participants	Each person's share_paise
expense_lines	Optional veg/grocery lines: "Tomato ₹40", "Bhindi ₹60"
settlements	from_member, to_member, amount_paise, status claimed approved rejected, claimed_at, decided_at, note
master_list	A view over expenses + settlements: when, who, type, what, price, status
shopping_items / inventory	Need vs have
milk_participation	Who drinks milk
devices	member_id + home Wi-Fi MAC / label for the beacon
current_presence	at_flat / away / unknown + last_seen_on_wifi. No GPS.
absence_periods	I'm home 20 Aug–10 Sep
keys / chores / issues / flat_info	Simple ops
recurring + rent_cycles	Wi-Fi / rent / electricity

Type chips: milk | vegetables | groceries | food | household | rent | electricity | wifi | other. Monthly report groups by these. Bills on the home UI can roll rent + electricity + wifi together.

Balance: paid – share + approved settlements received – approved settlements sent. Claimed-but-not-approved does *not* change the balance.

F. API — few actions

Action	Does
select_member	This phone is Ankit (cookie + local remember)
create_expense	Add veg / milk / anything → appears on master list
list_master	Filter by date, person, type
get_balances	Who owes whom (approved settlements only)
monthly_report	Month totals by type + per person
claim_settlement	I paid Rahul ₹X → waiting
approve_settlement / reject_settlement	Only the receiver can decide
ingest_wifi_presence	Beacon or app ping → At Flat
register_device	Save this phone's home Wi-Fi identity

Share math still runs on the server so four phones cannot disagree on ₹1000 / 3. That is arithmetic, not “security product.”

G. Who you are — not a login product

Screen

Who are you?

Ankit Joshi

Jayash Ghelot

Rahul Verma

Lakshit Pareek

Tap once. This phone stays you until you switch.

What we dropped

- Google Sign-In
- Apple Sign-In
- Email / magic link
- Invite tokens / OTP
- Passwords per person

Optional only — default off

If the website URL ever leaks outside the four phones, add one shared 4-digit flat code before the name list. You said keep it easy, so we will not build this unless you ask.

H. UI / UX

Mobile-first. Bottom nav: **Home · Money · Flat · People.**

Nav	Inside
Home	Who is home, balances, veg/milk peek, pending "Rahul please approve", due alerts, today's chore
Money	+ Add, Master list, Balances, Settlements (waiting / approved), Monthly report, Recurring / Rent
Flat	Shopping, Inventory, Milk, Vegetables, Keys, Chores, Issues, Info
People	Four faces, presence, balance, key, chores, my devices (Wi-Fi)

+ **Add** is one sheet with Type chips on top. Vegetables and Milk can also have a one-tap shortcut from Home / Flat. Master list is the diary. Monthly report is a real V1 screen, not a future slide.

I. Simple rules — not a bank

Only these four people will use it. We will not build OAuth, App Review, rate-limit dashboards, or invite RBAC. We still keep money honest.

Do	Don't
Tap name + remember this phone	Google / Apple / magic link
Rahul must approve money he received	Auto-clear debt because Ankit typed "paid"
Keep the link on your four phones	Enterprise roles and audit SSO
Integer paise; void with a note if someone typed wrong	Silent delete
Home Wi-Fi presence	Store GPS trails "for security"

J. Stack

Layer	Choice
App	Next.js + TypeScript, mobile-first PWA
Database	Postgres (Supabase as hosted Postgres is fine)
Auth	None of the usual providers. Name picker + simple session cookie.
Realtime	So Rahul sees Ankit's veg add and the "please approve" card without refresh
Home beacon	Tiny Python or Node script on any always-on device on flat Wi-Fi, posting "device seen"
Hosting	Vercel (site) + Supabase/Postgres (data)

K. Development phases

Phase	Build	Done when
0	This revised plan approved	No V3, tap-name, Wi-Fi presence, approval settlements, veg + master list + monthly report in V1
1	App shell + four name buttons + remember phone + seed four members	Each person opens and is themselves
2	Add with types (veg/milk/...), master list, balances, claim/approve settlement	Ankit's ₹180 sabzi shows for everyone; Rahul must approve a payment
3	Monthly report + recurring / rent / electricity	August page answers "kitna kharch hua"
4	Home dashboard + shopping, inventory, keys, chores, issues	Daily ops without WhatsApp
5	Home Wi-Fi beacon + register phone + presence on Home/People	2/4 home when two phones are on flat Wi-Fi

Ready to code when you say

1. Tap-name login is enough (no Google).
2. V3 stays deleted; monthly report + veg + master list + approve-settlement are V1.
3. Presence = home Wi-Fi beacon, not GPS.
4. Keep it simple for four people.

Shared Flat OS — Architecture v1.1 (revised) · 10 August 2026 · Four members only · Replaces v1.0.